Module 7 Assignment

1. Schema and evidence. Why these node and edge types, and what evidence you put on the edges.

I constructed an ovarian cancer (OV) knowledge graph containing 55 nodes and 62 edges. I chose six node types, including disease, gene, phenotype, pathway, drug, and variant, to integrate related genetic and pharmacological information. This map contains 15 ovarian cancer candidate genes, 10 HPO phenotypes, 6 Reactome pathways, drug-target relationships, genetic variation and a related disease.

The graph uses typed edges to represent gene–disease associations, disease–phenotype relationships, gene–pathway participation, drug–target interactions, and variant–gene mappings. These original data and evidence were from target_ov.csv, which was done in module 2. Also, I used GWAS catalog, HPO, Reactome, pharmacological database, and NCBI dbSNP to complete the target_ov_final.csv, which contains all the data. The gene-disease relationship contains available GWAS p values and source labels, while other relationships are supported by database annotations and reference materials. However, not all relationships have confidence scores. In addition, these associations cannot prove the causal relationship or that a certain drug can treat ovarian cancer.

2. Top targets. Which genes rank highest by centrality, and does that match the biology you expect? Flag any non-obvious hit.

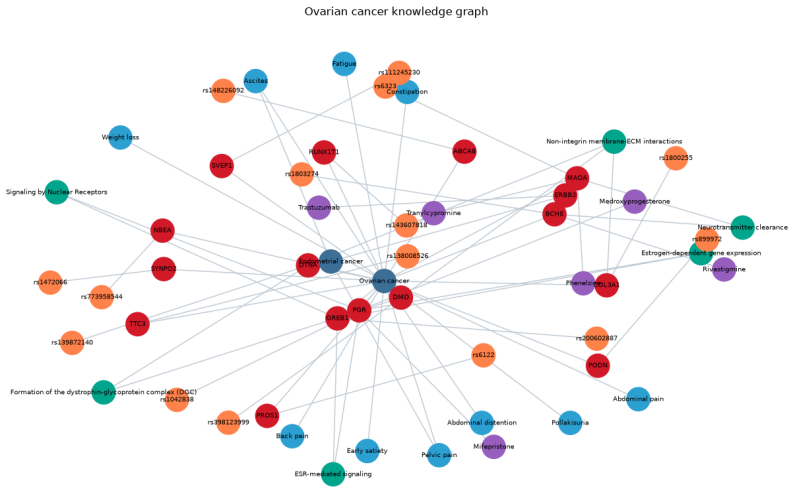
From step 5 in the notebook, based on PageRank centrality, PGR and MAOA are ranked highest among the genes in my ovarian cancer knowledge graph. Their two genes scored 0.0295, followed by BCHE, which is 0.0225. PGR is a result that meets biological expectations, because the progesterone receptor signal is related to the hormone-related process in ovarian cancer. In my graph, PGR is connected to three Reactome pathways and two drugs. The three pathways are Estrogen-dependent gene expression, ESR-mediated signaling, and Signaling by Nuclear Receptors. In addition, it also connects two drugs through a drug-target relationship, which are Mifepristone and Medroxyprogesterone. From these connections, it means that PGR links several biological pathways with pharmacological information, making it an interesting and useful candidate for further research. However,

although the PGR has a high PageRank score ,which shows its position in the current knowledge map, it does not mean that it is the most important driving gene for ovarian cancer.

MAOA and BCHE are relatively unexpected findings because their role in ovarian cancer has not been fully established compared with their known biological functions. In my knowledge graph, both genes are connected to the pathway named Neurotransmitter clearance. In addition, MAOA is linked with Phenelzine and Tranylcypromine, these two drugs, while BCHE is connected with Rivastigmine.

These connections show that both genes have relationships with biological pathways and existing drug targets, which may help explain why they have a relatively high PageRank score. However, their ranking may partially reflect the structure of my knowledge graph and the drug-target relationship it contains, but it does not necessarily mean that there is sufficient evidence to show that these genes can drive ovarian cancer. This point I think it's important because drugs linked with MAOA and BCHE are mainly used for other indications, and they cannot be directly considered to be treated as ovarian cancer treatment drugs.

In contrast, DMD and SYNPO2 have GWAS p-value in my data set, but their PageRank scores are lower, while the other genes don't have. This difference shows that PageRank reflects the connection between genes and other nodes in the knowledge map, while the GWAS p value reflects the statistical significance of the genetic association. Therefore, genes with high PageRank scores are not necessarily more closely related to ovarian cancer. When selecting targets which are worthy for further research, the connection relationship in the network and genetic evidence should be considered at the same time.



3. A hypothesis from a path. State one repurposing or shared-mechanism hypothesis a path in your graph suggests, and say what would test it.

About the hypothesis from a path, maybe my ovarian cancer knowledge graph can put forward a hypothesis of drug repurposing based on the path from Mifepristone to PGR and then to ovarian cancer. Mifepristone is a drug that acts on progesterone receptors(Autry & Wadhwa, 2024), and PGR is the gene that encodes this receptor, and it is also one of the candidate genes I selected through multi-omics analysis. Additionally, my analysis results also connect PGR with hormone-related signaling pathways, including ESR-mediated signaling and Signaling by Nuclear Receptors. Based on these connections, a hypothesis can be put forward that the drug, Mifepristone, targeting PGR may affect hormone-related biological processes in ovarian cancer cells.

However, this is just the hypothesis according to the knowledge graph and it cannot prove that Mifepristone is the effective drug to cure ovarian cancer.

To verify this hypothesis, first, I will use ovarian cancer cell lines with different PGR expression levels and then use different concentrations of Mifepristone to treat these cells. Also, I will compare them with the control group that did not receive drug treatment. For the result, I will observe how the growth of cells, living percentage and apoptosis change. Plus, I will test the changes of relevant signal pathways to judge whether this drug influences the biological process mentioned in the knowledge graph. To make sure whether the effect is related to PGR, I will use the gene knockdown method to reduce the expression of PGR, and then observe whether the response of cells to Mifepristone changes. If the experiment finds Mifepristone can produce repeatable effects, and this effect is related to the expression of PGR, it can support further research on whether Mifepristone has the potential for repurposing of ovarian cancer drugs.

4. An edge is evidence, not truth. Which of your edges are weak, and how would the answer change with better data? Note where you used (or would use) AI to harmonize data or write queries, and how you checked it.

In my ovarian knowledge graph, there are some connections that are weaker than others. For example, in the GWAS data set of ovarian cancer I downloaded, only DMD and SYNPO2 matched the relevant records, while the other 13 candidate genes did not match the GWAS p

value. But it does not mean that other genes are not related to ovarian cancer, because they were initially selected based on multi-omics evidence. Also, the drug-target relationship shows that drugs cannot prove that these drugs can treat ovarian cancer. Similarly, the mutation-gene relationship means that genetic mutations are mapped to certain genes, which does not prove that these mutations lead to ovarian cancer. These limitations show that the knowledge map should be understood as the integration of different biological evidence, not the proof of causality. If I get more complete GWAS data and validated drug-target relationships, I can evaluate these connections more reliably. This additional evidence may change the priority ranking of genes and help identify ovarian cancer targets with more research potential.

I used AI to help integrate ovarian cancer GWAS data with my list of candidate genes and Ensembl identifiers. I verified the results by checking the merged target_ov_ final.csv file, confirming that all 15 genes were retained, and also checking that DMD and SYNPO2 matched the GWAS record.

In addition, I used Reactome Pathway Analysis to identify the biological pathways associated with candidate genes. I entered the list of genes, checked the results of the pathway enrichment analysis, and checked which genes correspond to each selected pathway.

Reference:

Autry, B. M., & Wadhwa, R. (2024, February 28). *Mifepristone*. StatPearls Publishing.

PubMed. https://www.ncbi.nlm.nih.gov/books/NBK557612/